

Extending a Diophantine quadruple

Problèmes ouverts — A. Dujella · 2026

Problem 1.1 (Dujella, 2026). Find all extensions of the rational Diophantine quadruple

$$\left\{ \frac{1}{16}, \frac{33}{16}, \frac{17}{4}, \frac{105}{16} \right\}$$

to a rational Diophantine quintuple.

Remark. Known extensions are $\frac{549120}{10201}$, $-\frac{26880}{177241}$. Is there any other possibility?